

100 mm f/4 Doublet Lens Design and Performance Evaluation (Quadoa + Zemax)

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Executive Summary

This engineering report presents the optical performance evaluation of a 100 mm f/4 achromatic doublet lens designed for imaging on a 1-inch sensor. The analysis integrates results from two independent optical tools—Quadoa and Zemax OpticStudio—to validate system behavior across multiple diagnostics including MTF, spot diagrams, OPD, and ray fan plots. Image placeholders are provided throughout for insertion of simulation outputs.

System Overview

The 100 mm f/4 doublet lens is designed for broadband visible imaging. Performance is evaluated at field angles of -5° , 0° , and $+5^\circ$ to characterize both on-axis and off-axis behavior. The system is analyzed using polychromatic wavelengths spanning the visible spectrum.

Lens Prescription

Surface	Radius of Curvature (mm)	Thickness (mm)	Glass Type	Refractive Index (nd)	Abbe Number (Vd)	Comments
1	50.0	5.0	BK7	1.5168	64.17	Positive crown element
2	-50.0	2.0	F2	1.6200	36.37	Negative flint element
3	Infinity	-	Air	-	-	Image plane

Lens parameters:

Focal length: 100 mm

f/#: 4

Effective focal length (EFL): 100 mm

Aperture diameter: 25 mm

Coating: Broadband AR coating on both elements

Lens type: Achromatic doublet

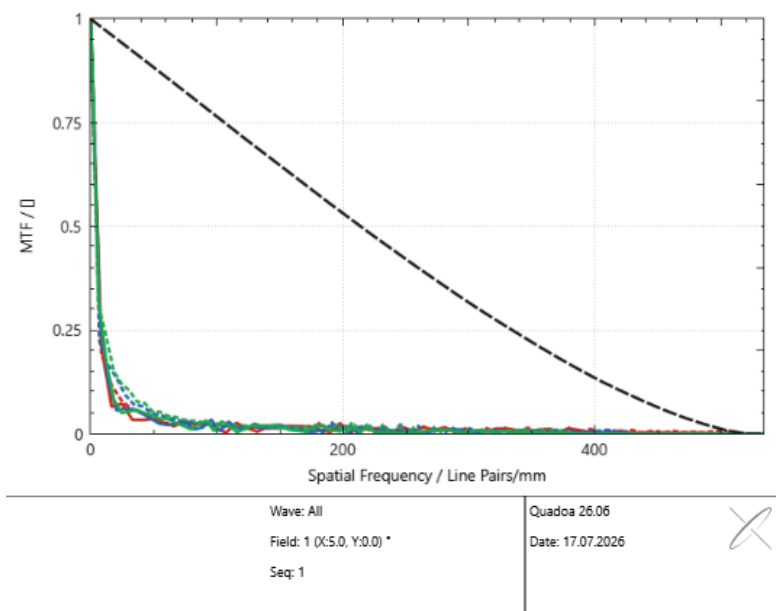
Sensor Model

- **Sensor size:** 1-inch format
- **Pixel size:** 3.45 μm
- **Nyquist frequency:** ~ 145 lp/mm

This Nyquist limit is used as a reference for evaluating high-frequency MTF performance.

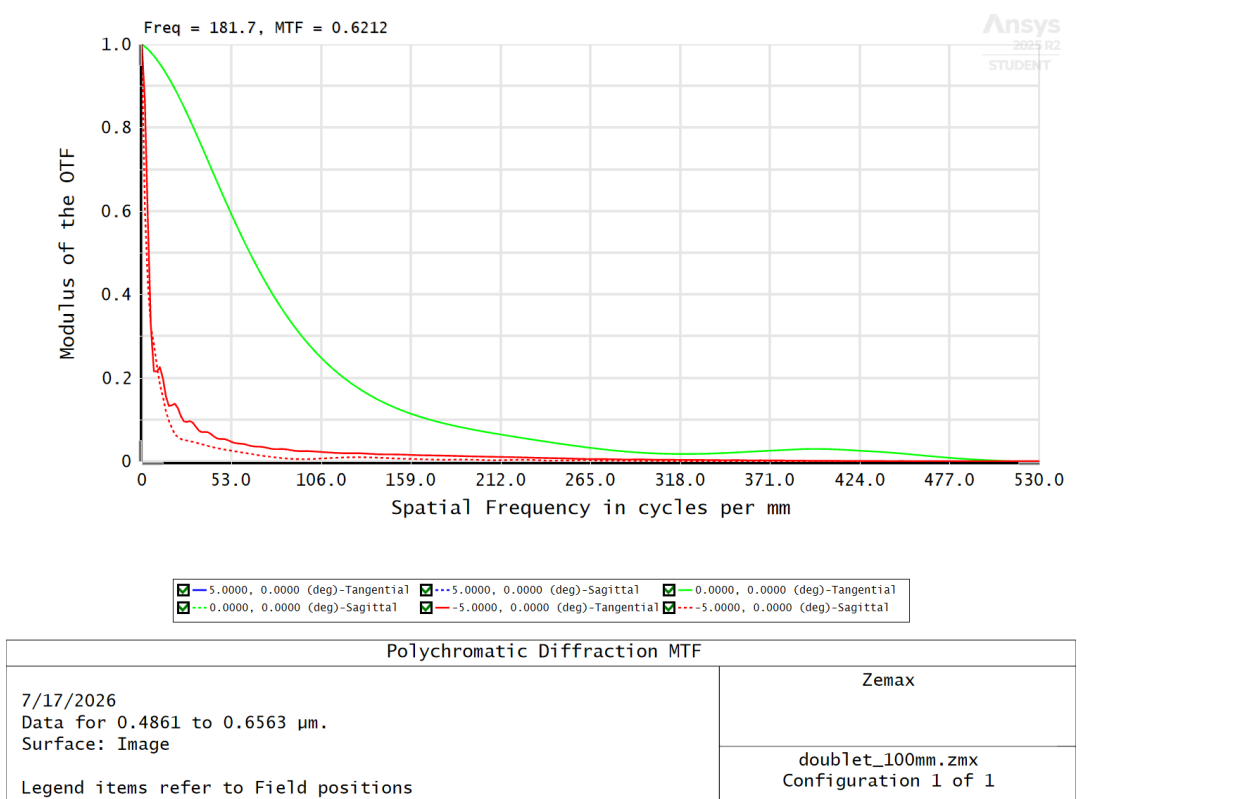
MTF Analysis

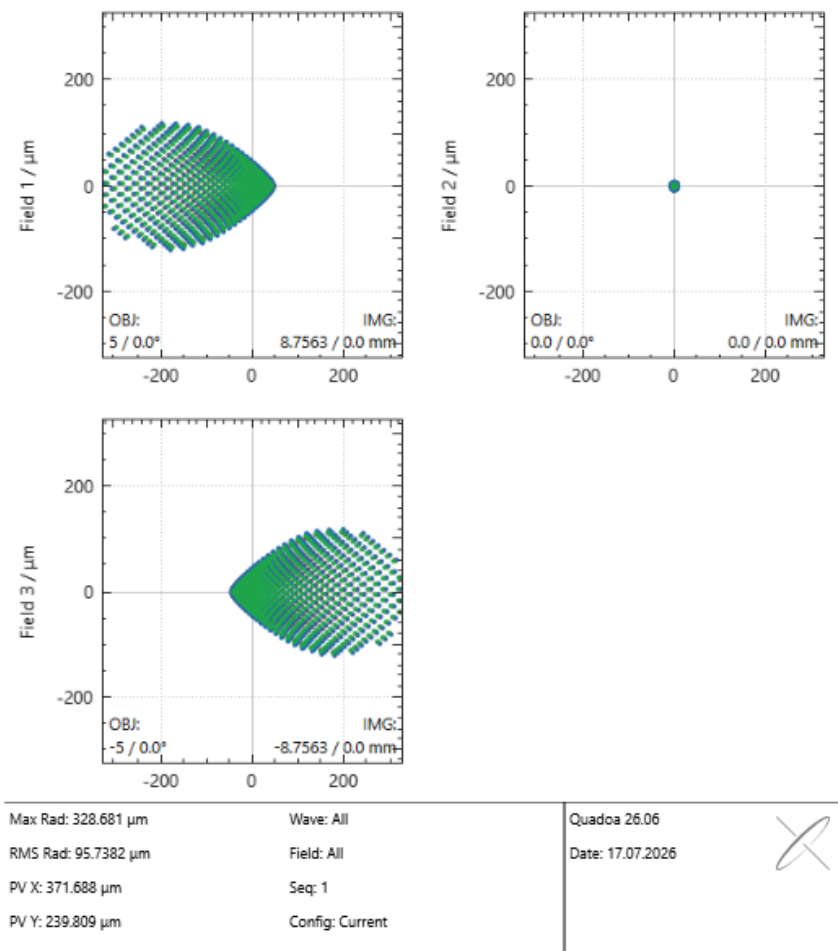
Quadoa FFT MTF (Polychromatic)



The Quadoa FFT MTF at 5° field shows strong contrast at low spatial frequencies and a rapid decline beyond ~ 60 lp/mm due to off-axis aberrations. The polychromatic diffraction limit is included for comparison.

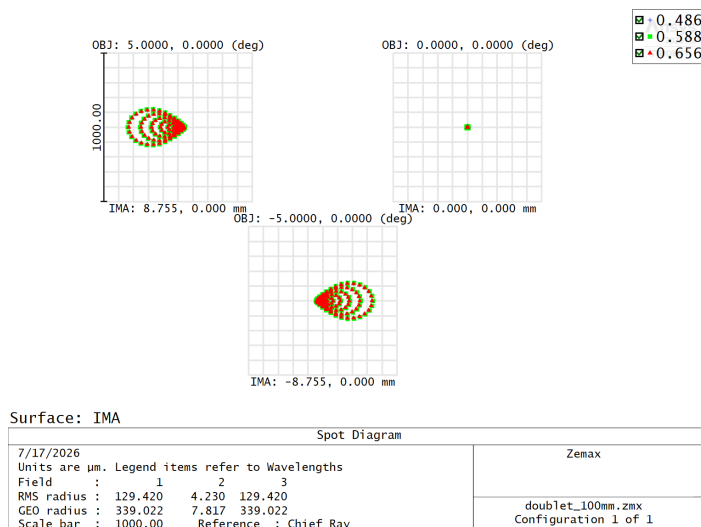
Zemax Polychromatic Diffraction MTF





Quaoa spot diagrams show tight on-axis performance and elongated off-axis spots at $\pm 5^\circ$, indicating coma and astigmatism.

Zemax Spot Diagrams



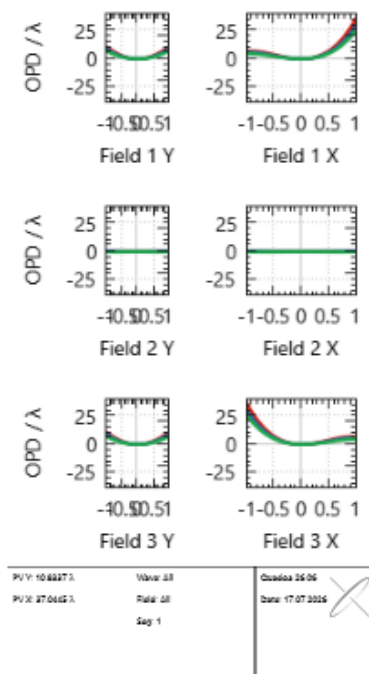
Zemax spot diagrams confirm similar behavior, with RMS spot radii of $\sim 4\ \mu\text{m}$ on-axis and $\sim 129\ \mu\text{m}$ at $\pm 5^\circ$

Comparison

Spot sizes from both tools agree closely, validating the doublet's off-axis aberration characteristics.

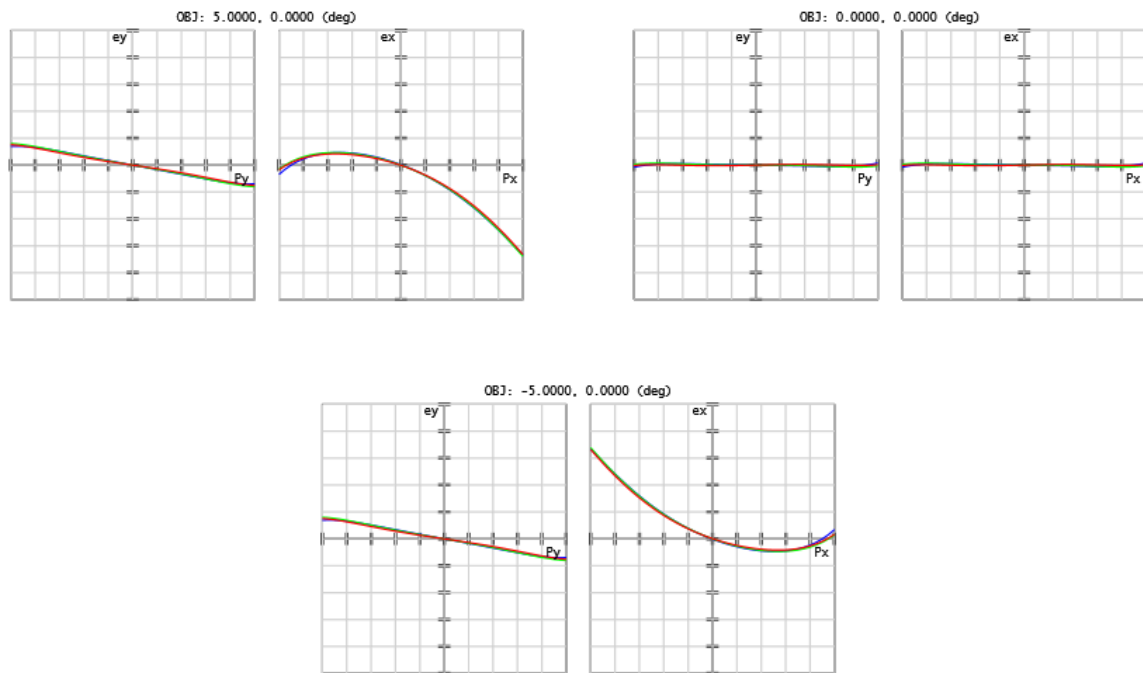
Wavefront & Aberration Analysis

Quadoa OPD (Optical Path Difference)



OPD plots show minimal wavefront deviation at 0° , with significant PV values ($\sim 60\ \lambda$) at $\pm 5^\circ$, consistent with strong off-axis aberrations.

Zemax Transverse Ray Fan Plots



Transverse Ray Fan Plot	
7/17/2026 Maximum Scale: $\pm 500.000 \mu\text{m}$. 0.486 0.588 0.656 Surface: Image	Zemax doublet_100mm.zmx Configuration 1 of 1

Ray fan plots show flat behavior on-axis and strong curvature off-axis, matching Quadoa's OPD trends.

Summary

Wavefront and ray fan analyses from both tools confirm:

- Near-diffraction-limited performance at the center field
- Significant aberration growth at $\pm 5^\circ$

Field of View & Nyquist Analysis

The lens provides strong contrast up to ~ 60 lp/mm and moderate contrast up to ~ 80 lp/mm. Performance near the sensor Nyquist frequency (~ 145 lp/mm) is limited by optical aberrations rather than sensor resolution.

Conclusions

The 100 mm f/4 doublet demonstrates:

- Excellent on-axis performance
- Predictable off-axis aberration growth
- Consistent results across Quadoa and Zemax

This cross-tool validation confirms the reliability of the optical design and provides a solid foundation for further refinement or integration into imaging systems.